

AI Fundamentals

Module 1

Collaborate with AI

Module 2

Practice using AI

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Tell us about your experience with AI in under 2 minutes

Ungraded (Pledge) • 15 min

●

Get the most out of Lala

Reading • 4 min

●

Complete labs with Google AI Pay at no cost

Reading • 4 min

●

Walken your Google AI Pro trial

Ungraded (Pay later) • 4 min

●

Brainstorm ideas with Gemini

Labs • 12 min

●

Can AI have been blowing your mind?

Review? Video • 12 sec

Module 3

Learn how AI works

●

Learn foundational AI concepts

Video • 2 min

●

Fundamentals of machine learning

Reading • 4 min

●

Understand AI limitations

Reading • 8 min

●

Leverage AI features and capabilities for work

Video • 4 min

Module 4

Design great prompts

Module 5

Level up your prompts

Module 6

Use AI responsibly

Understand AI limitations

0:11 / 8:08

A key part of being a responsible AI user is knowing how AI works and where it can fall short. AI has limitations, such as potential biases, knowledge cutoffs, and drift in output. Recognizing these limitations can help you evaluate outputs more effectively and use AI in a way that's fair, accurate, and responsible.

Understand biases in AI

Data bias can be a foundational challenge for AI. This can happen when the data used to train an AI model is skewed, incomplete, or reflects historical or societal biases. Because the model's output is directed by its training data, it can sometimes reproduce existing biases in its response, like connecting specific activities with certain age groups.

Part of following a responsible AI approach is being aware that AI models may create stereotypes or bias in their outputs. For example, if you ask AI to generate an image of an office, it might consistently produce an image of a high-rise building. This may occur because its training data comes from urban business hubs. As a result, it may struggle to generate images of creative workspaces, home offices, or rural business settings. Your role is to guide the model toward a fair and impartial output.

To avoid further bias in outputs, you can use these techniques in your prompts:

- **Be specific about the output you want:** Add important context about your intended audience and their needs. You can also provide fair and balanced references for the model to follow.
- **Use follow-up prompts to correct outputs that seem biased or inaccurate:** If AI provides a biased response, point out the stereotype when you iterate on your prompt and ask the model to correct the bias.

The constraint of knowledge cutoff

Knowledge cutoff is the point in time when a model's training data ends. This means the model lacks information on events, discoveries, or data that occurred after that date.

You might notice that a model can provide information about very recent events. Some may do this by performing a live web search to find current information that they can supplement their answer with. It's helpful to think of this as the difference between what the model knows from its training versus what it can look up in the moment. The model's core knowledge is not continuously updated, which is why the concept of a knowledge cutoff remains a critical limitation to bear in mind.

Responsible AI use requires you to verify time-sensitive information. Always use a search engine or other reliable sources to fact-check statistics, news, or any information about recent events.

To work effectively with a model's knowledge cutoff, you can use these techniques:

- **Look up the cutoff:** You can search online for the knowledge cutoff date of a specific AI tool. This helps you understand the boundaries of its internal knowledge.
- **Verify time-sensitive information:** For any statistics, breaking news, or details about recent events, always cross-reference AI's answer with a reliable external source, like a search engine or an official report.
- **Specify your timeframe:** When asking about a topic that changes over time, state the timeframe for what you need. For example, instead of "What was the biggest song of the summer?" ask "What was the biggest song of the summer in 2023?"
- **Refine with follow-up prompts:** If an answer seems outdated, like calling a product new when it is several years old, use a follow-up prompt to ask for more recent alternatives or clarification.

Changes in AI's performance over time

Drift is the gradual decline in a model's accuracy and relevance as the real world changes. You might observe drift in two ways:

- **Factual drift:** This is when AI becomes less accurate over time because of its knowledge cutoff. For example, as AI models' advice on current fashion trends may become less useful the further you get from its training date.
- **Behavioral drift:** This refers to changes in AI's behavior over time. As developers update models, you might notice that the formatting, tone, or conversational style changes, even when you use the same prompts.

Here are a few ways to manage and mitigate both kinds of drift:

- **Provide accurate and up-to-date context in your prompts,** especially for topics that change quickly, like market trends or technology.
- **Keep chats focused by starting a new conversation for each specific task.** This also helps to reset the context window if a conversation becomes too long or the output starts to feel off topic.
- **Be explicit** with clear and specific instructions in your prompts.

AI is powerful, which is why it's important to use it carefully and responsibly. That means being mindful of AI's limitations and accounting for biases. The best results come from a combination of human ingenuity and AI's capabilities. So keep asking insightful questions, refining your prompts, and above all, applying your critical thinking skills to verify outputs before you use them. That way, you contribute to a future where AI serves everyone responsibly.

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